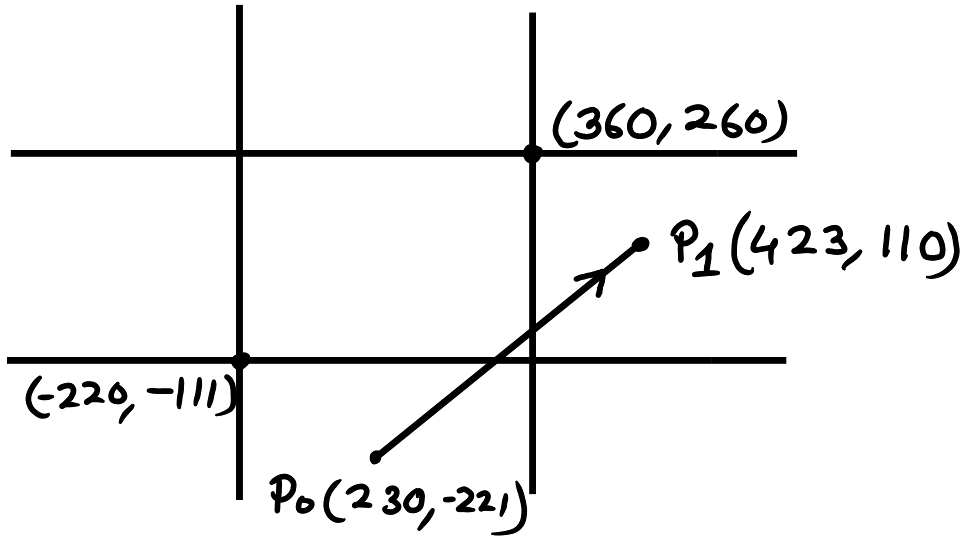
Set A

Q1.



$$\begin{array}{lcccc} & T & B & R & L \\ P_0 \approx OC1 = & 0 & 1 & 0 & 0 \\ P_1 \approx OC2 = & 0 & 0 & 1 & 0 \\ \hline AND = & 0 & 0 & 0 & 0 \end{array}$$

P_0: $\left\{ \begin{array}{l} x_{min} < x_0 < x_{max} \\ y_0 < y_{min} \end{array} \right.$

P_1: $\left\{ \begin{array}{l} x_1 > x_{max} \\ y_{min} < y_1 < y_{max} \end{array} \right.$

$\therefore$ Partially inside.

Cohen-Sutherland

$$m = \frac{110+221}{423-230} = 1.72$$

P_0, $0\overset{\substack{\text{Bottom} \\ \downarrow}}{1}00$, $y = y_{min} = -111$

$$m = \frac{y_{min} - y_0}{x - x_0}$$

$$\Rightarrow x \approx 293.95$$

$\therefore P_0'(293.95, -111) \approx OC1' = 0000$

P_0, $\overset{\text{Right}}{\overset{\downarrow}{0010}}$, $x = x_{max} = 360$

$$m = \frac{y - y_1}{x_{max} - x_1}$$

$$\Rightarrow y \approx 1.64$$

$\therefore P_1'(360, 1.64) \approx OC1' = 0000$

So, Clipped points: $(293.95, -111), (360, 1.64)$

Cyrus-Beck

[if anyone answered using Cyrus-Beck]

$$\bar{D} = (423 - 230, 110 + 221)$$
$$= (193, 331)$$

$t_E = 0$, $t_L = 1$

Boundary	N_i	$N_i \cdot D$	t	PE/PL	t_E	t_L
L	$(-1, 0)$	$-193 < 0$	$\frac{-(230+220)}{(423-230)}$ $= -2.33 < 0$ ✗	PE	0	1
R	$(1, 0)$	$193 > 0$	$\frac{-(230-360)}{(423-230)}$ $= 0.67 < 1$ ✓	PL	0	0.67
B	$(0, -1)$	$-331 < 0$	$\frac{-(-221+111)}{110+221}$ $= 0.33 > 0$ ✓	PE	0.33	0.67
T	$(0, 1)$	$331 > 0$	$\frac{-(-221-260)}{110+221}$ $= 1.45 > 0.67$ ✗	PL	0.33	0.67

$\therefore P_0'(0.33) = P_0 + 0.33\,D \approx (293.69, -111.77)$

$\therefore P_1'(0.67) = P_0 + 0.67\,D \approx (359.31, 1.95)$

Q2.

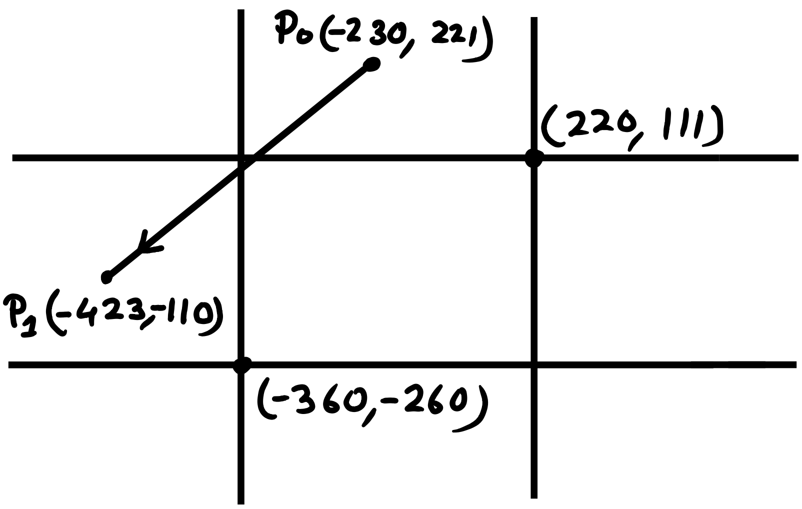
$$P = S_c^{-1} \times R_f^{-1} \times R^{-1} \times P'$$

$$= \begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{3} & 0 & 0 \\ 0 & 0 & \frac{1}{1} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(30^\circ) & -\sin(30^\circ) & 0 & 0 \\ \sin(30^\circ) & \cos(30^\circ) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 5 \\ 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3.08 \\ 3.11 \\ 0 \\ 1 \end{bmatrix} \quad \text{(Answer)}$$

Set B

Q1.



$$P_0 \approx OC1 = \begin{array}{cccc} T & B & R & L \\ 1 & 0 & 0 & 0 \end{array} \rightarrow \begin{cases} x_{min} < x_0 < x_{max} \\ y_0 > y_{max} \end{cases}$$

$$P_1 \approx OC2 = 0\ 0\ 0\ 1 \rightarrow \begin{cases} x_0 < x_{min} \\ y_{min} < y_0 < y_{max} \end{cases}$$

$$AND = 0\ 0\ 0\ 0$$

$\therefore$ Partially inside.

Cohen-Sutherland

$$m = \frac{-110-221}{-423+230} = 1.72$$

P_0, $\overset{\text{Top}}{1}000$, $\quad y = y_{max} = 111$

$$m = \frac{y_{max} - y_0}{x - x_0}$$

$$\Rightarrow x \approx -293.95$$

$\therefore P_0' (-293.95, 111) \approx OC1' = 0000$

P_0, 0001 (Left), $x = x_{min} = -360$

$$m = \frac{y - y_1}{x_{min} - x_1}$$

$$\Rightarrow y \approx -1.64$$

$\therefore P_1'(-360, -1.64) \approx OC1' = 0000$

So, Clipped points: $(63.95, 111)$, $(-360, -1.64)$

Cyrus-Beck [if anyone answered using Cyrus-Beck]

$$\bar{D} = (-423+230, -110-221)$$
$$= (-193, -331)$$

$t_E = 0$, $t_L = 1$

Boundary	N_i	$N_i \cdot D$	t	PE/PL	t_E	t_L
L	$(-1, 0)$	$193 > 0$	$\frac{-(-230+360)}{(-423+230)} = 0.67 < 1$ ✓	PL	0	0.67
R	$(1, 0)$	$-193 < 0$	$\frac{-(230-220)}{(423-230)} = -2.33 < 0$ ✗	PE	0	0.67
B	$(0, -1)$	$331 > 0$	$\frac{-(221+260)}{-110-221} = 1.45 > 0.67$ ✗	PL	0	0.67
T	$(0, 1)$	$-331 < 0$	$\frac{-(221-111)}{-110-221} = 0.33 > 0$ ✓	PE	0.33	0.67

$\therefore P_0'(0.33) = P_0 + 0.33\,D \approx (-294.14, 111)$

$\therefore P_1'(0.67) = P_0 + 0.67\,D \approx (-359.31, -1.95)$

Q2.

$$P = S_c^{-1} \times R_f^{-1} \times R^{-1} \times P'$$

$$= \begin{bmatrix} \frac{1}{1} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{3} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(60^\circ) & -\sin(60^\circ) & 0 & 0 \\ \sin(60^\circ) & \cos(60^\circ) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 5 \\ 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.67 \\ 5.58 \\ 0 \\ 1 \end{bmatrix}$$ (Answer)